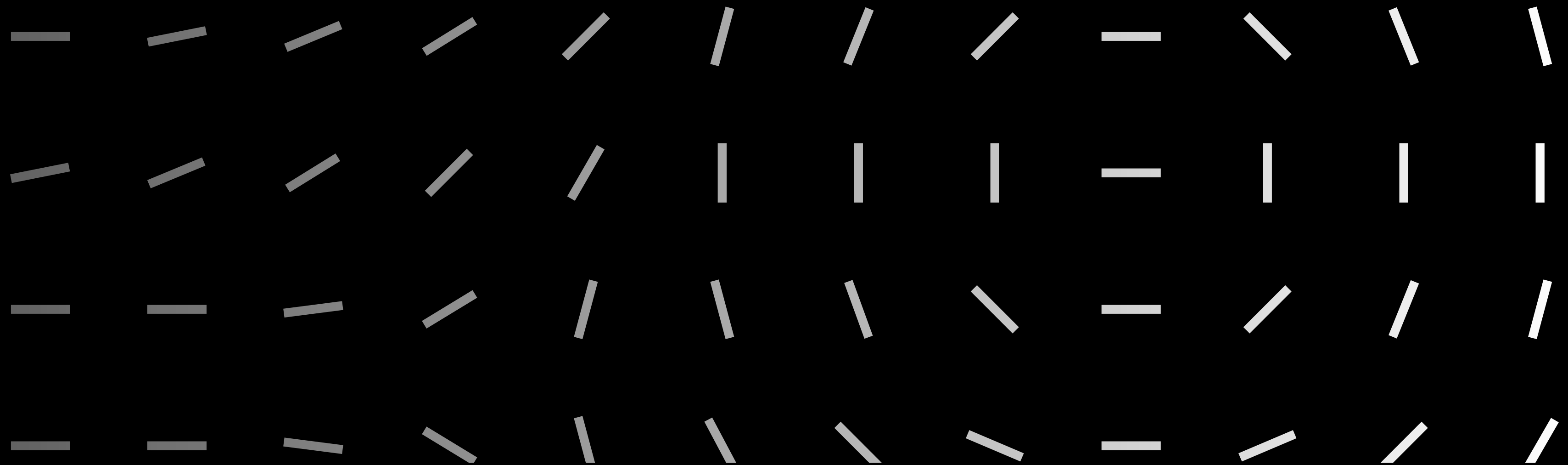




Agentic AI

Session 2

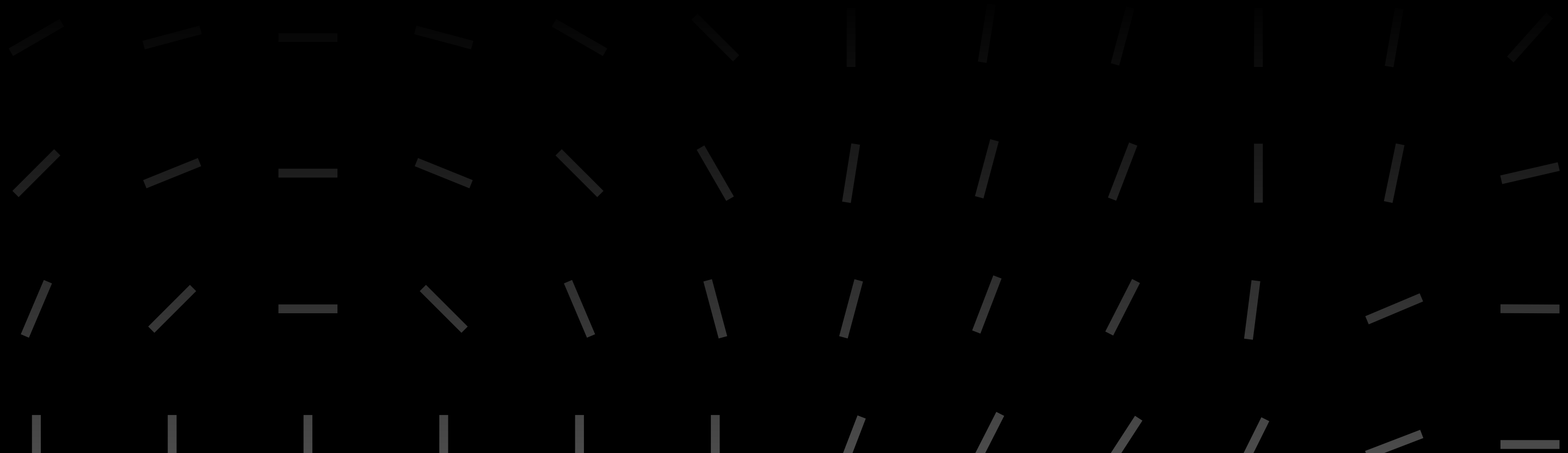
Presented by Eduard and Huy



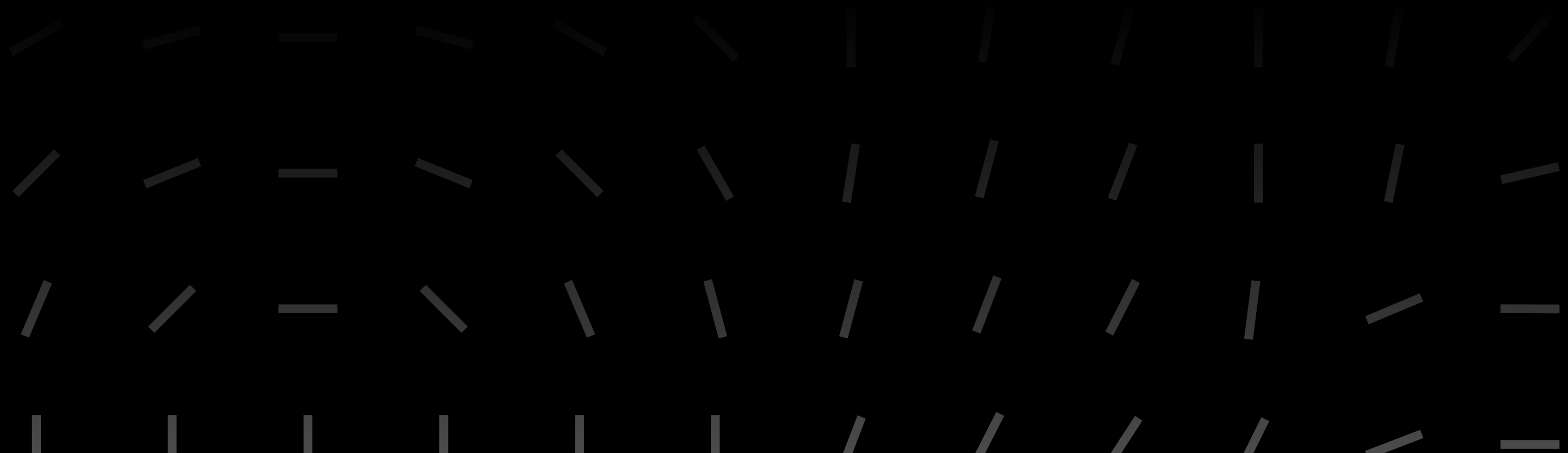
Session 2 Contents

- RAG
- Tool Calling
- Industry Tools

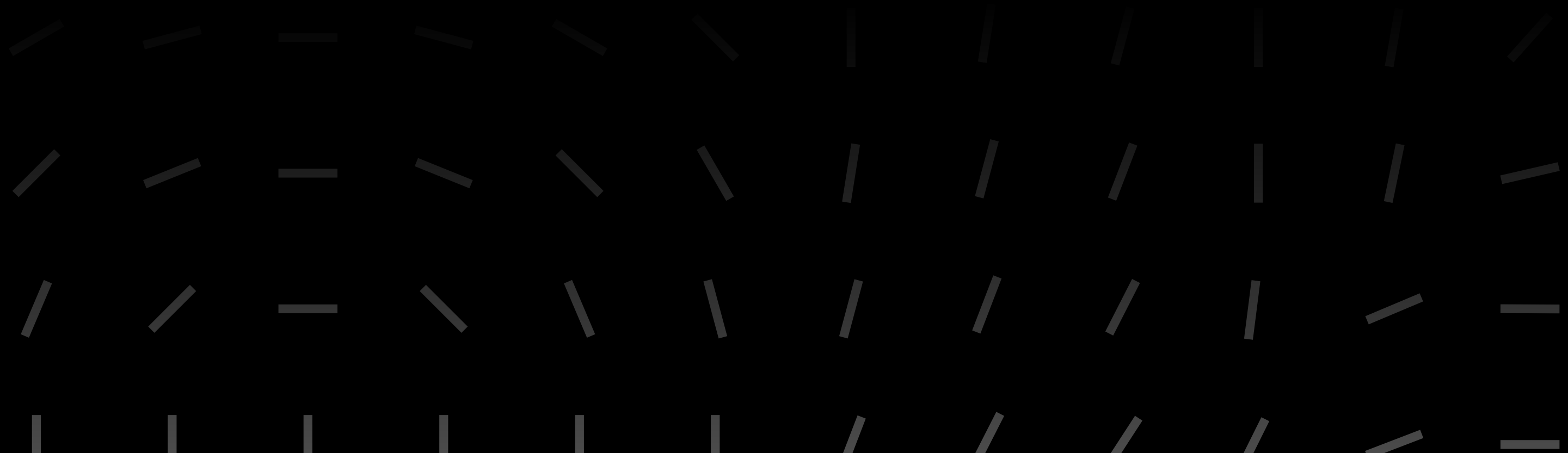
Before we jump in!

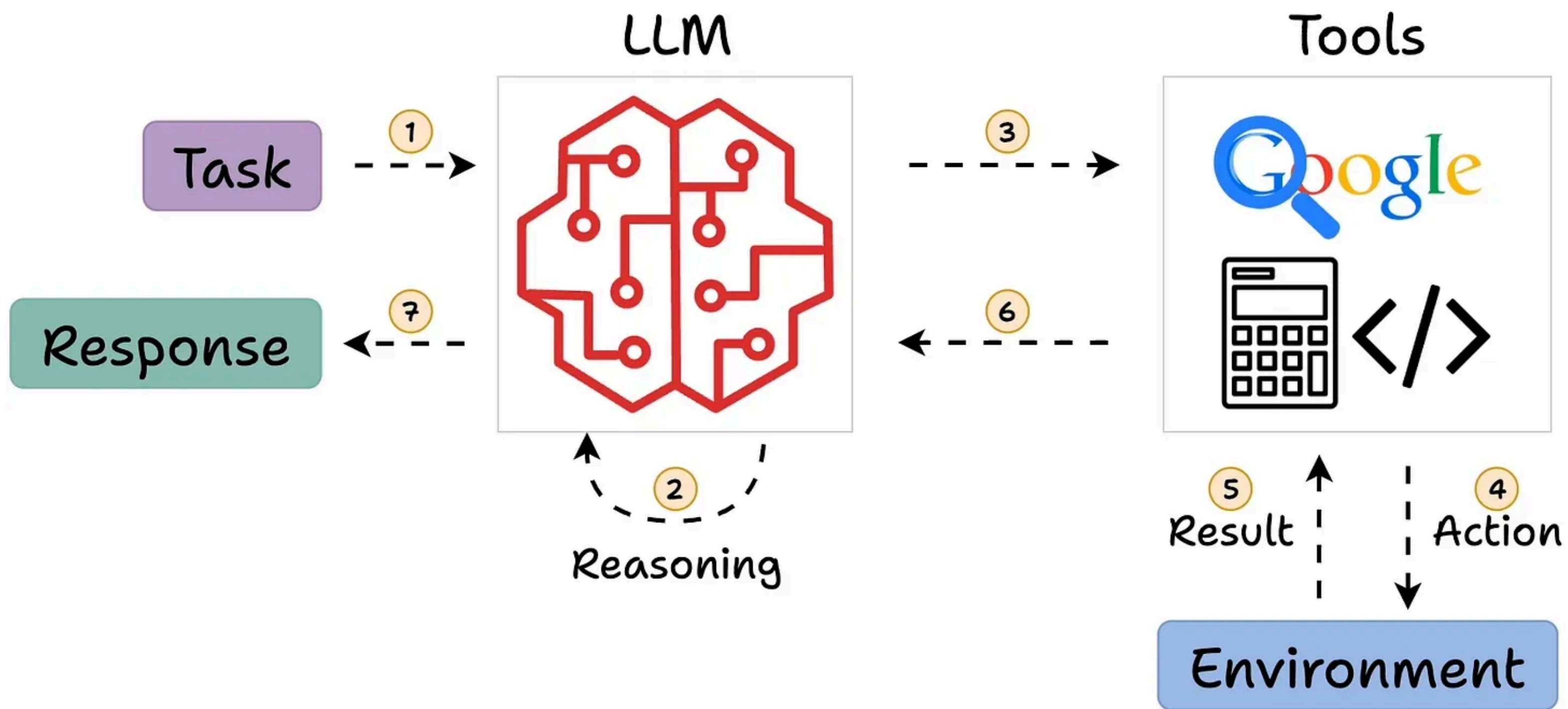


Tool Calling: Beyond just text



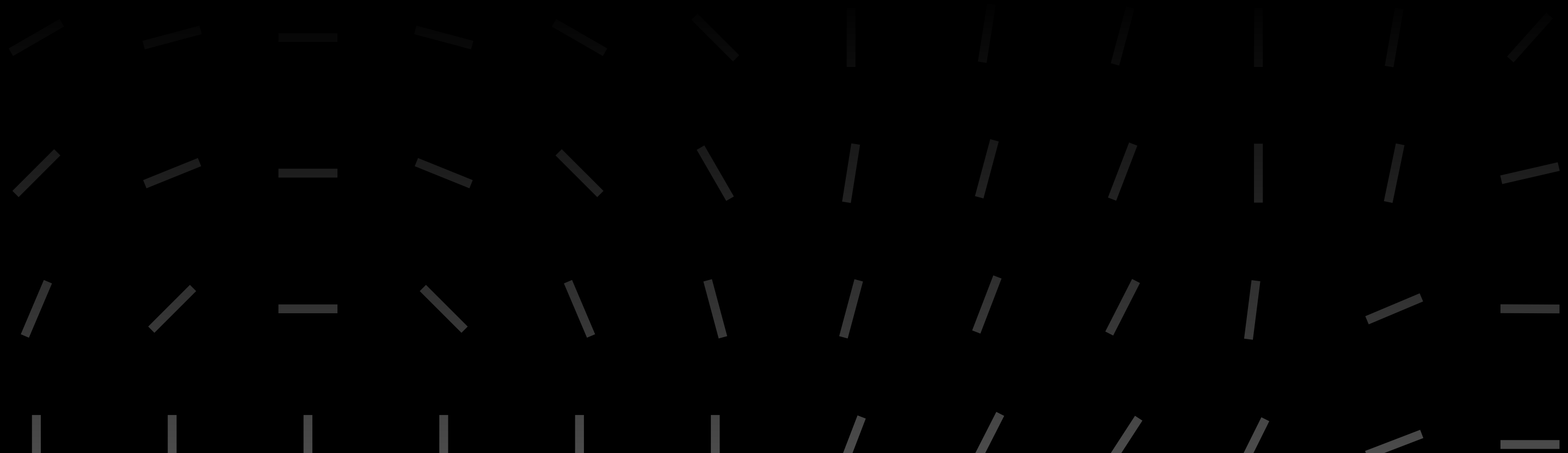
Tool Calling: How Does It Work



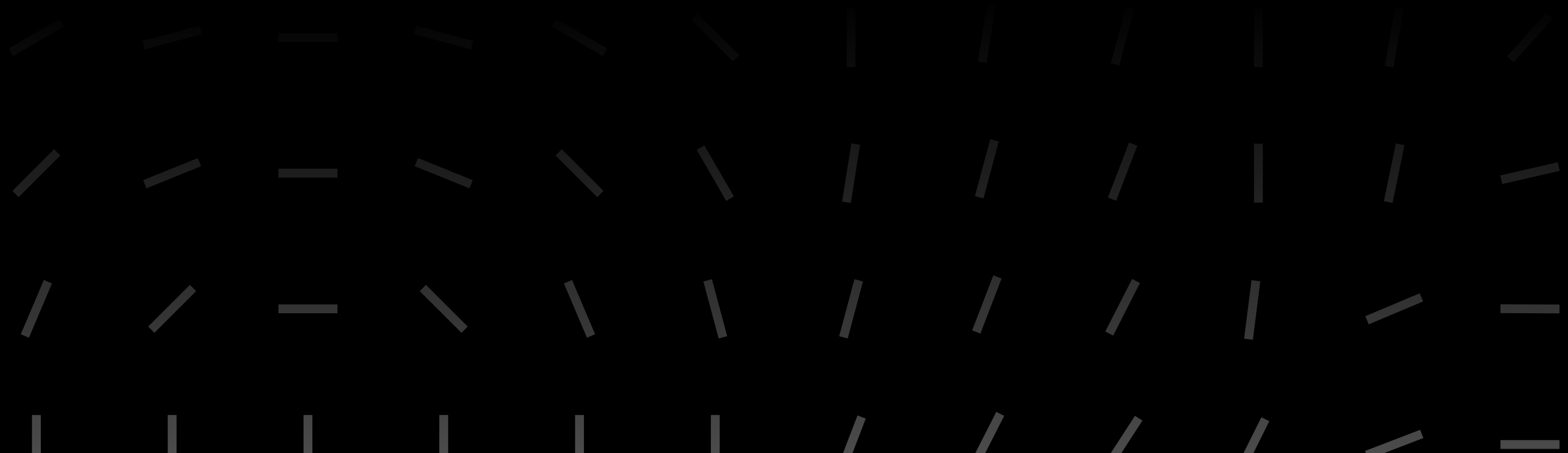


Tool calling DEMO

Questions?

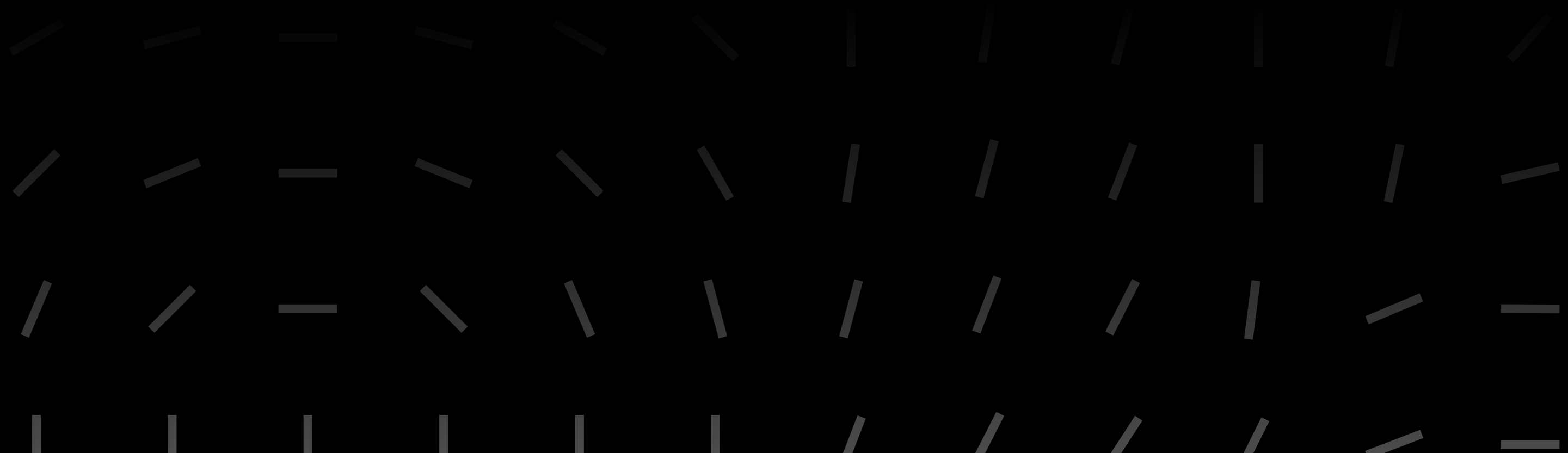


What is Retrieval-Augmented Generation (RAG)?

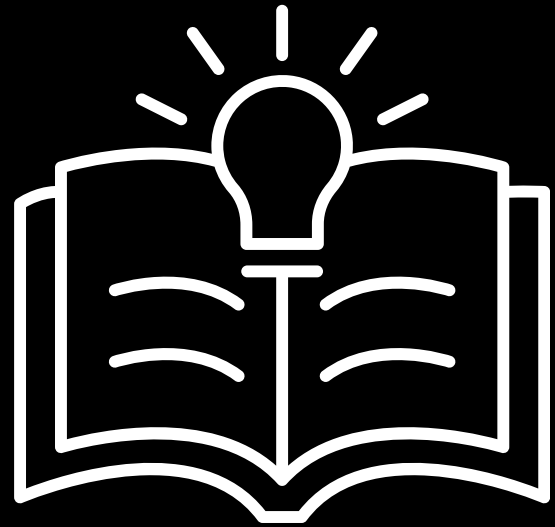


What is Retrieval-Augmented Generation?

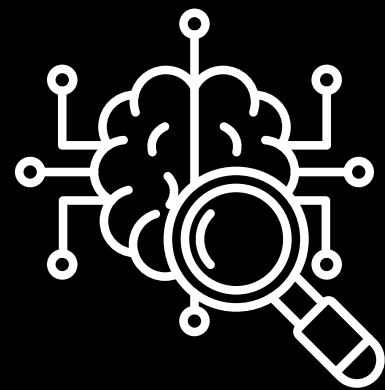
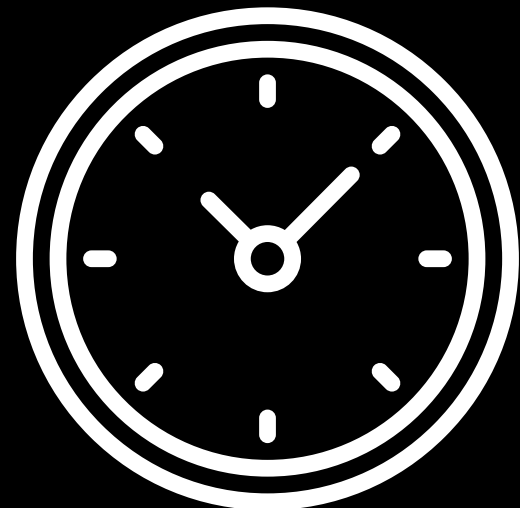
The Problems First!



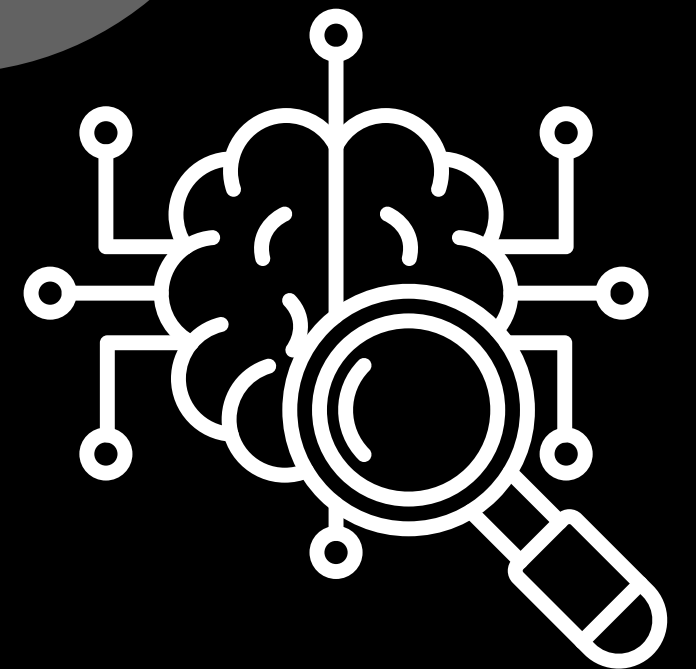
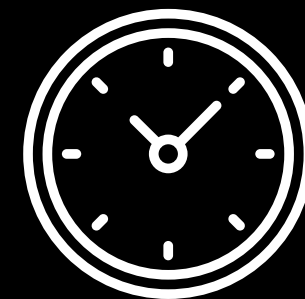
Large Language Model



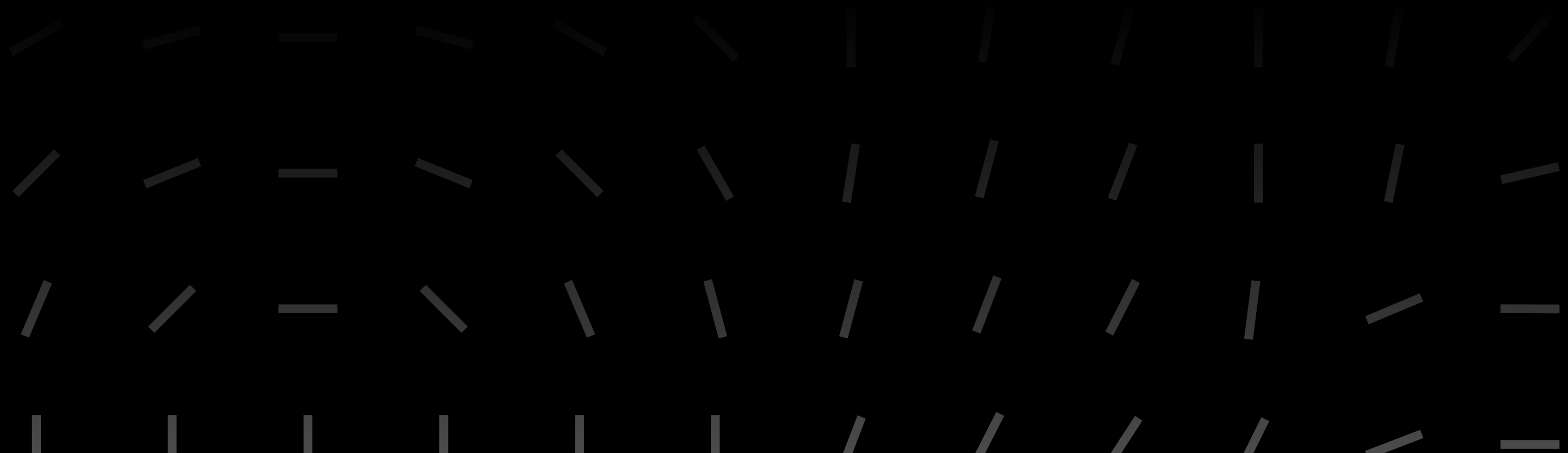
Base LLM



Instruct
LLM



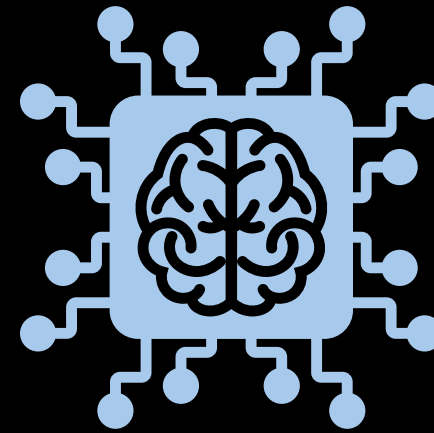
Questions?



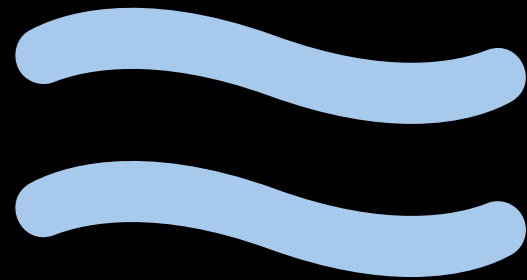
Retrieval



- vector store

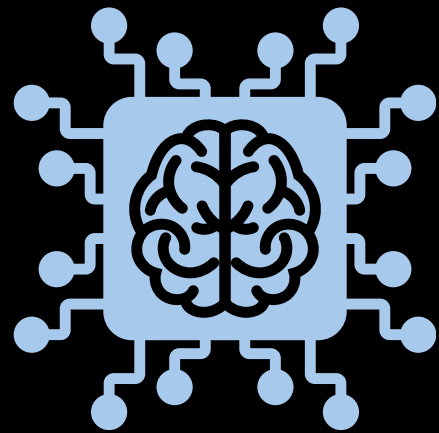


- embeddings model



- semantic
similarity search

Retrieval



- embeddings model

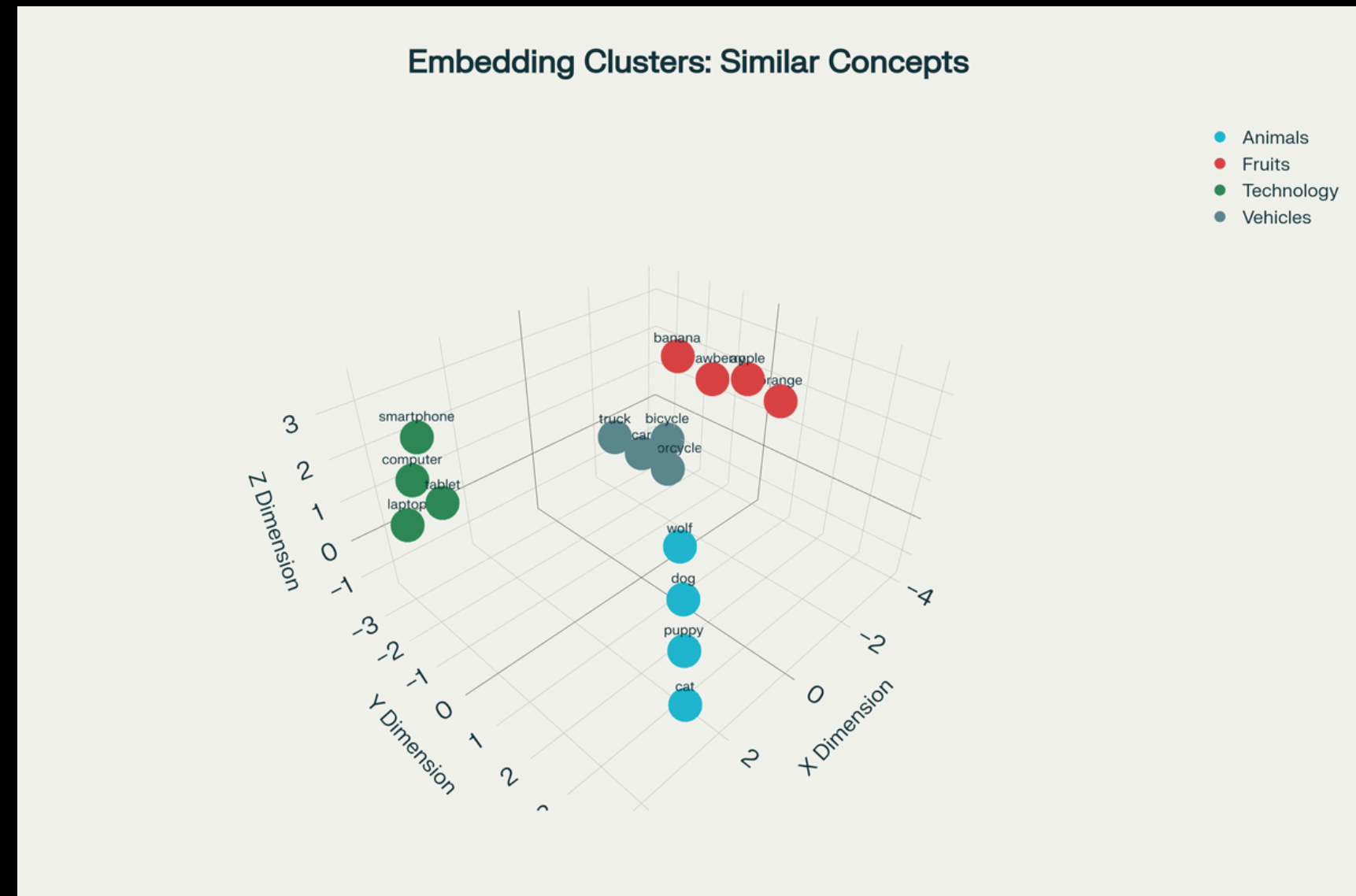


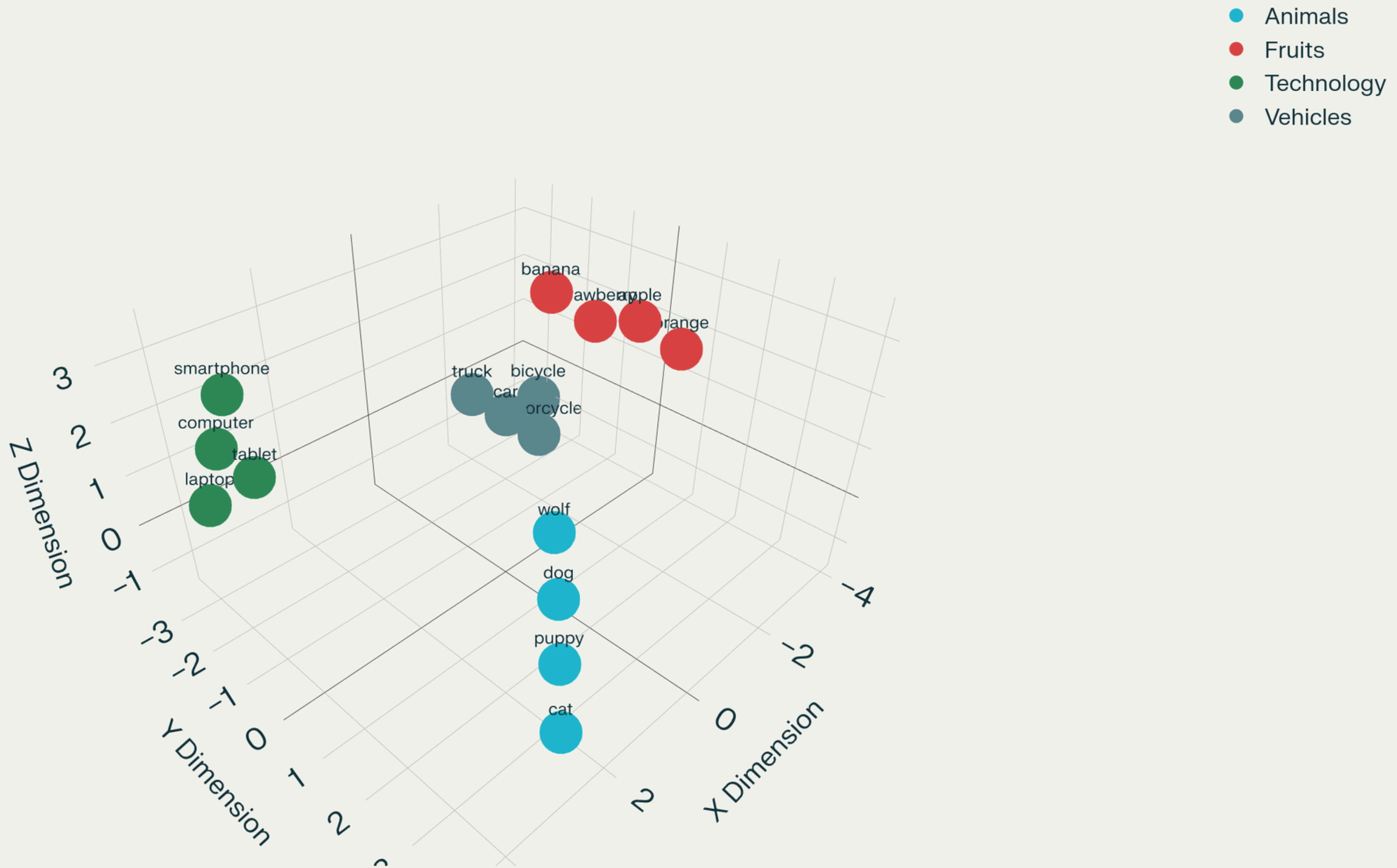
Gemini

Retrieval

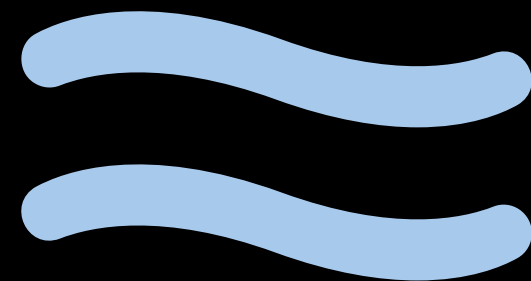


- vector store

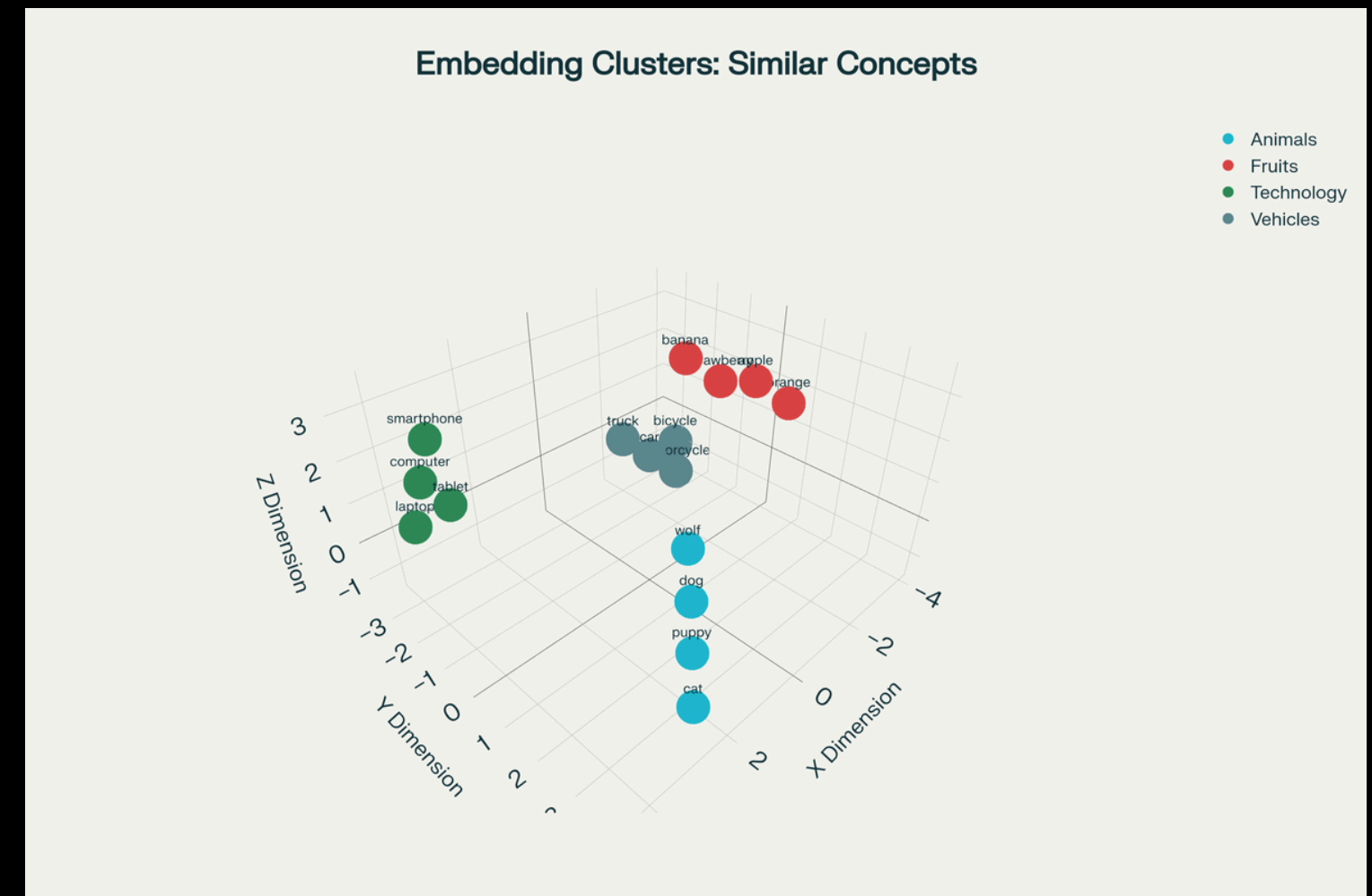




Retrieval



semantic similarity
search

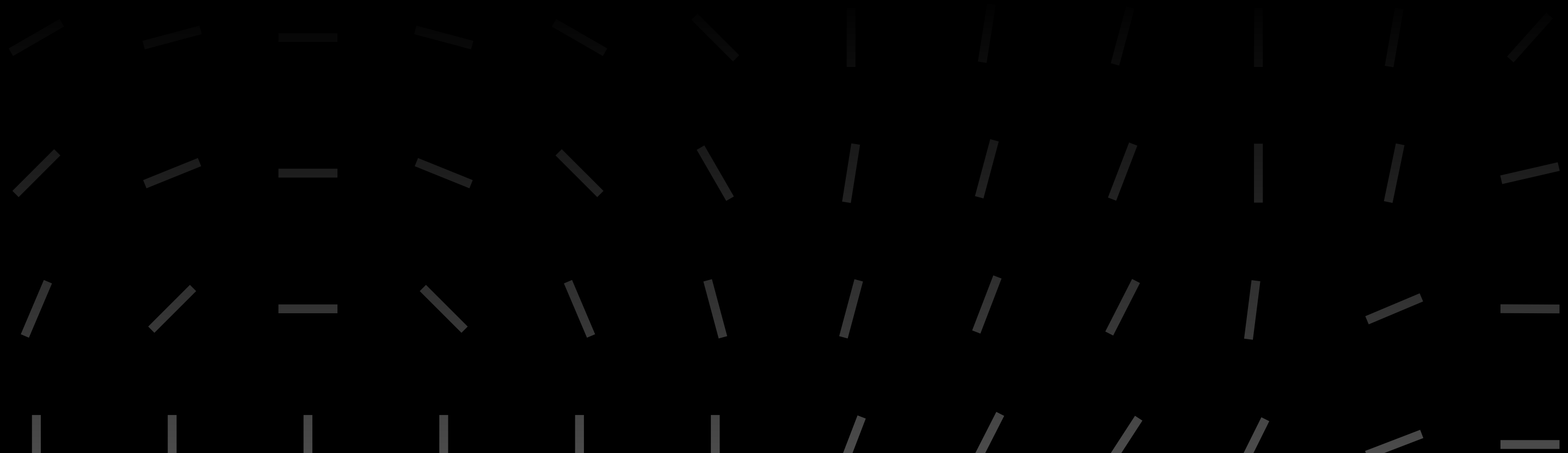


Retrieval Demo

Augmented

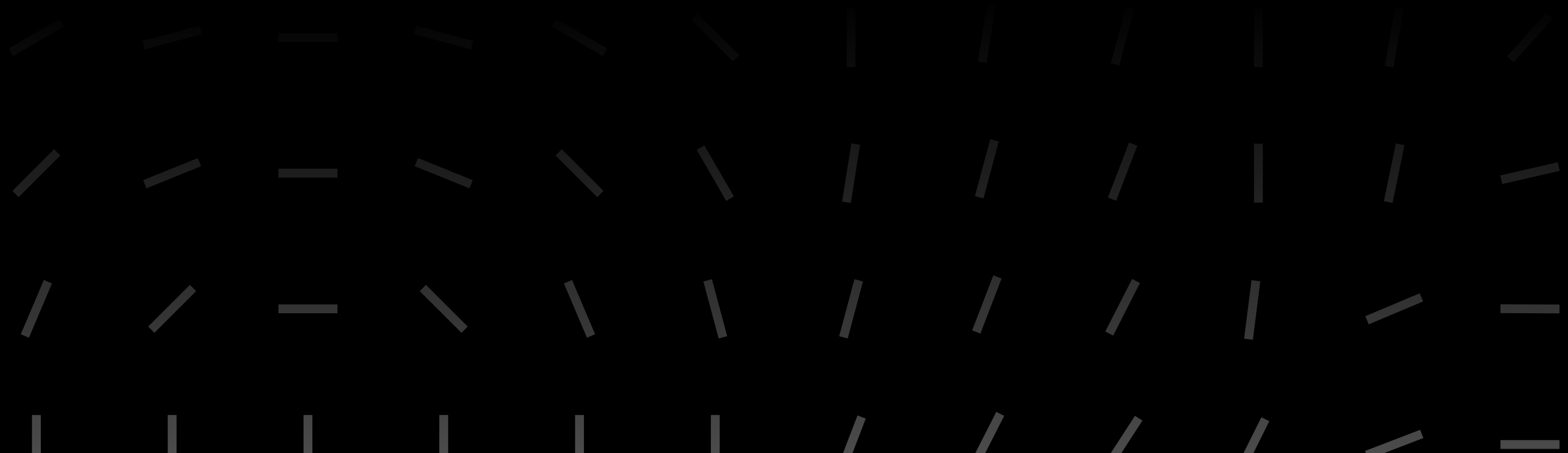


Generation

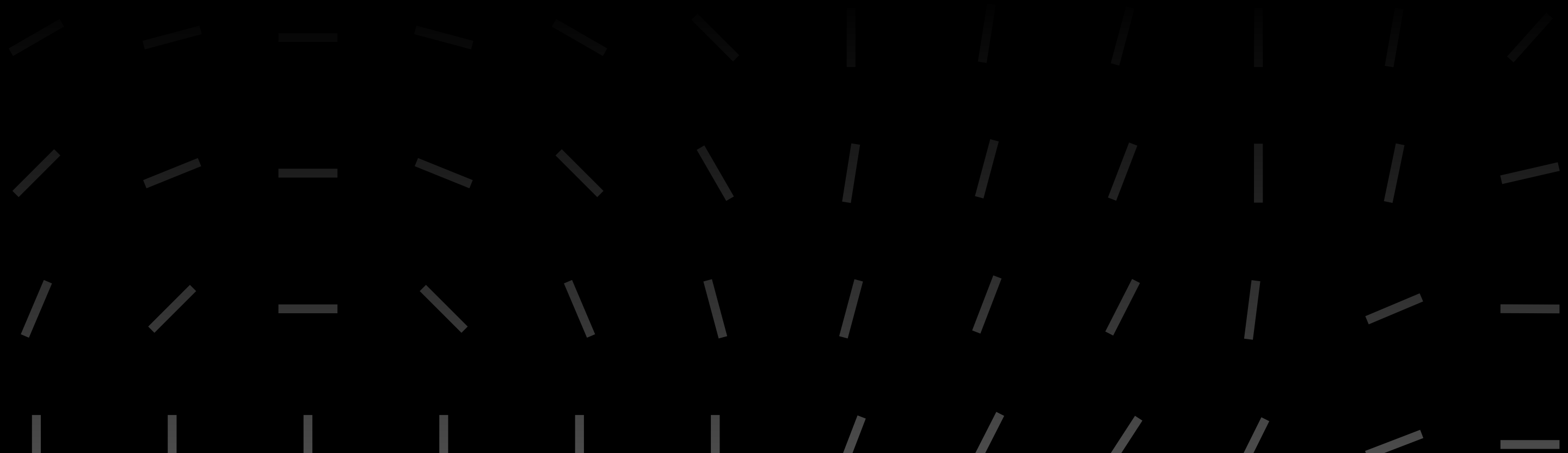


RAG Demo

Questions?



AI Apps We Use



Comprehensive Market Research with Perplexity



<https://pplx.ai/huy-huynh>

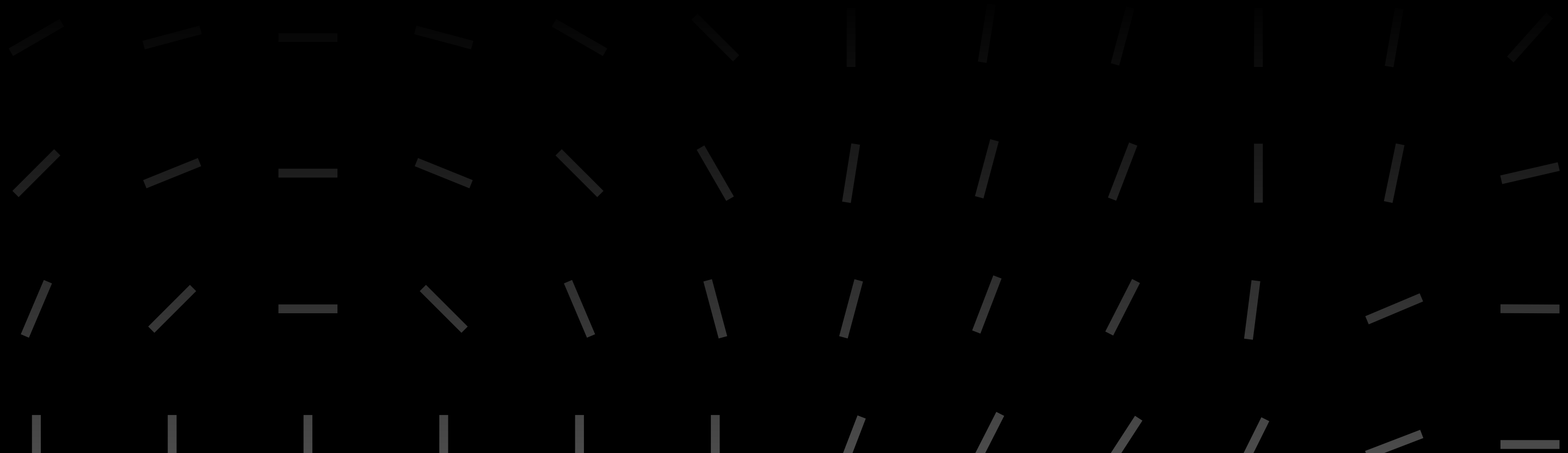
Build a simple Study Tracker website with Lovable



Listen to your study notes with NotebookLM



Let's install n8n



Steps

1. Navigate

<https://nodejs.org/en/download>

Download Node.js®

Get Node.js® v22.19.0 (LTS) for macOS using nvm with npm

Info Want new features sooner? Get the [latest Node.js version](#) instead and try the latest improvements!

```
1 # Download and install nvm:
2 curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.3/install.sh | bash
3
4 # in lieu of restarting the shell
5 \. "$HOME/.nvm/nvm.sh"
6
7 # Download and install Node.js:
8 nvm install 22
9
10 # Verify the Node.js version:
11 node -v # Should print "v22.19.0".
12
13 # Verify npm version:
14 npm -v # Should print "10.9.3".
```

Bash Copy to clipboard

"nvm" is a cross-platform Node.js version manager. If you encounter any issues please visit [nvm's website](#)

Or get a prebuilt Node.js® for macOS running a ARM64 architecture.

macOS Installer (.pkg) Standalone Binary (.gz)

Read the [changelog](#) or [blog post](#) for this version.

Learn more about [Node.js releases](#), including the release schedule and LTS status.

Learn how to [verify](#) signed SHASUMS.

Looking for Node.js source? Download a signed [Node.js source](#) tarball.

Check out our [nightly](#) binaries or all [previous releases](#) or the [unofficial](#) binaries for other platforms.

Click

Steps

2. Paste into your terminal/command prompt and hit enter

Linux/Mac

- Search for terminal

Windows

- Search for Command prompt


```

% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed

100 16631  100 16631    0     0  122k      0 --:--:-- --:--:-- --:--:-- 122k
=> Downloading nvm from git to '/Users/eduardkakosyan/.nvm'
=> hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /Users/eduardkakosyan/.nvm/.git/
=> Compressing and cleaning up git repository

=> nvm source string already in /Users/eduardkakosyan/.zshrc
=> Appending bash_completion source string to /Users/eduardkakosyan/.zshrc
=> Close and reopen your terminal to start using nvm or run the following to use it now:

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh" # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion" # This loads nvm bash_completion
Downloading and installing node v22.19.0...
Downloading https://nodejs.org/dist/v22.19.0/node-v22.19.0-darwin-arm64.tar.xz...
##### 100.0%
Computing checksum with sha256sum
Checksums matched!
Now using node v22.19.0 (npm v10.9.3)
v22.19.0
10.9.3

```

Steps

3. Within the same terminal window type ``npm install -g n8n``

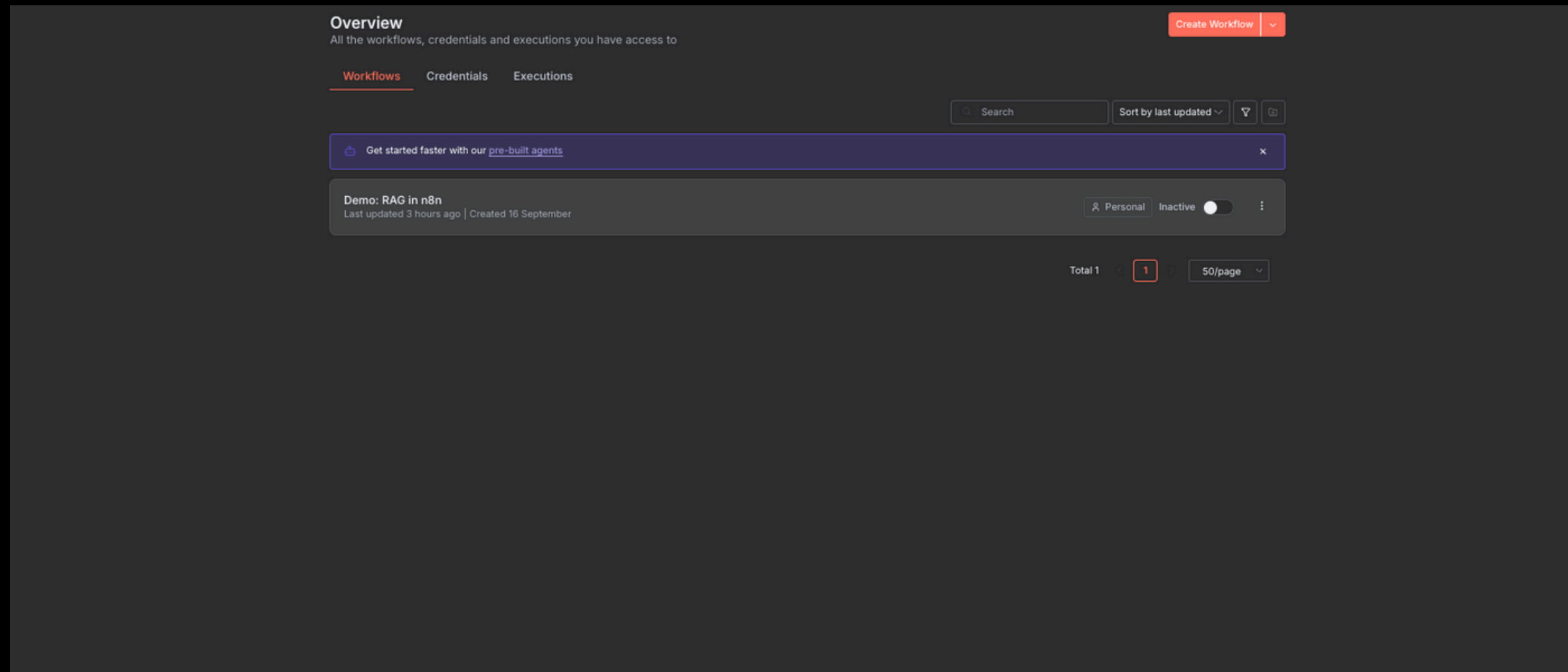
```
added 1965 packages in 1m
```

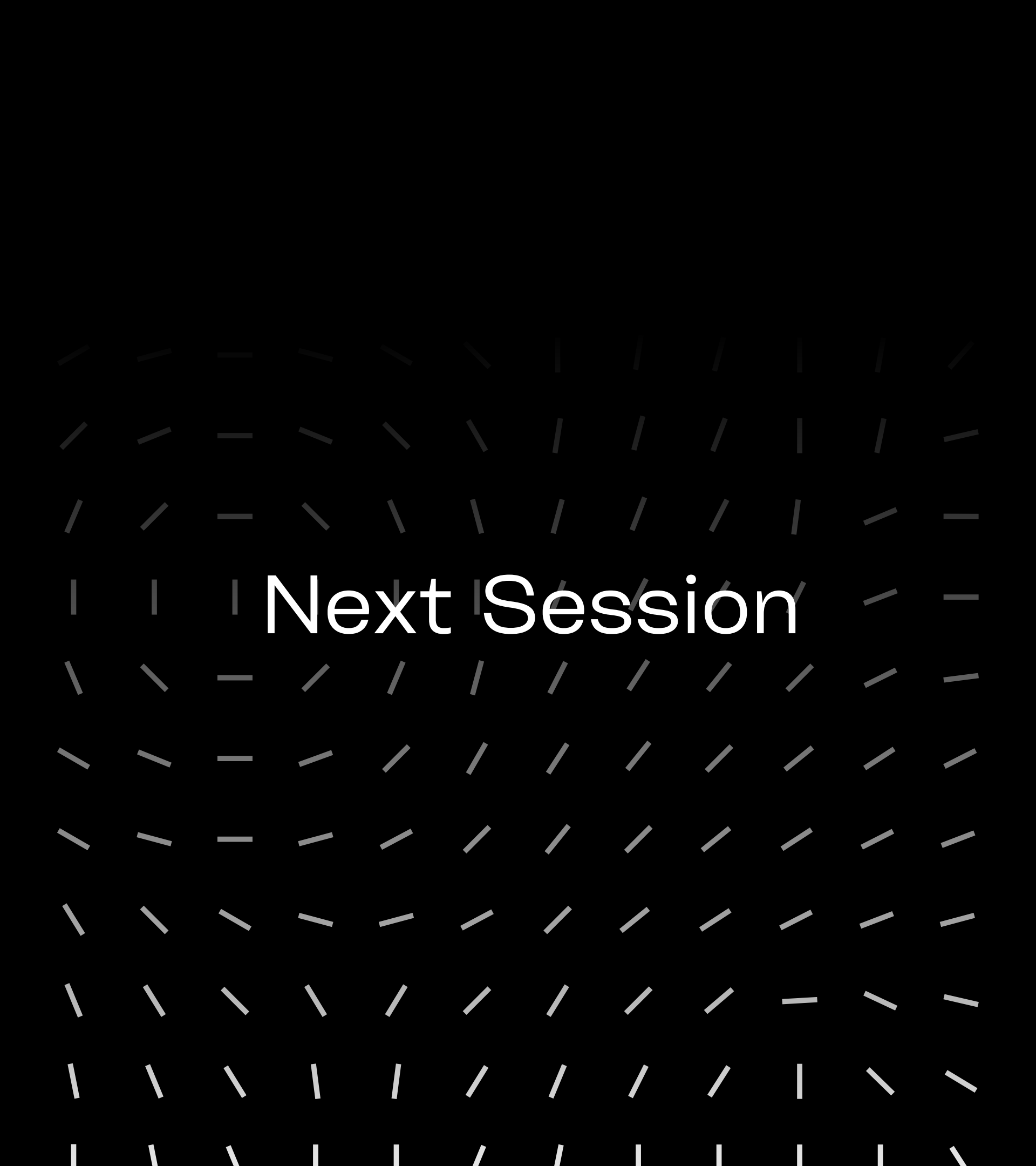
```
233 packages are looking for funding
```

```
run `npm fund` for details
```

Steps

4. Within the same terminal window type n8n and press “o”.
Create an account with your Dal email, and sign-in





Next Session

- Workflows
- Agents

Questions?

Ask us on LinkedIn



Eduard Kakosyan



Huy Huynh

